

国家中小学课程资源

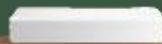
实验：电池电动势和内阻的测量（第一课时）

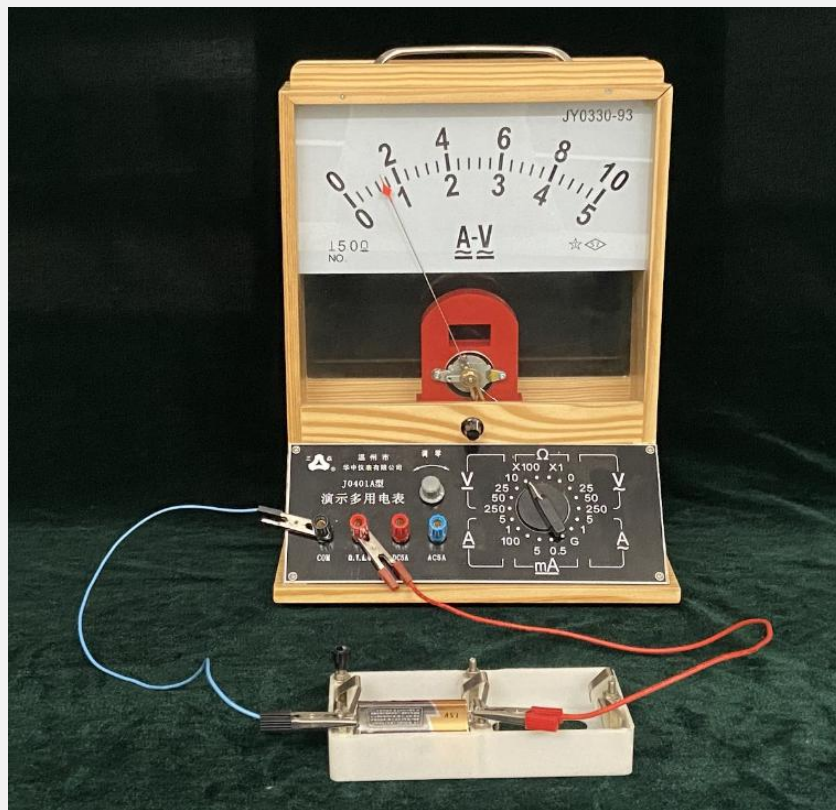
年 级：高二

学 科：物理（人教版）

主讲人：周艳

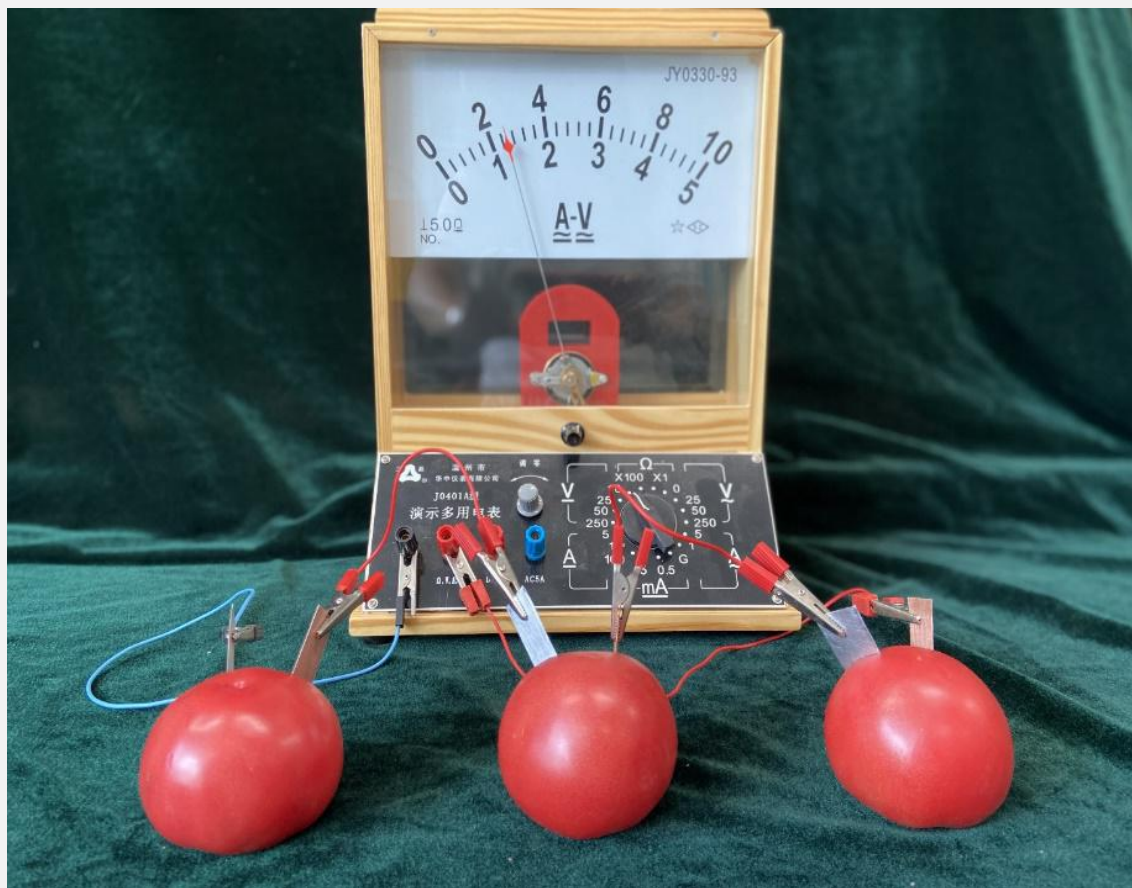
学 校：北京师范大学附属实验中学





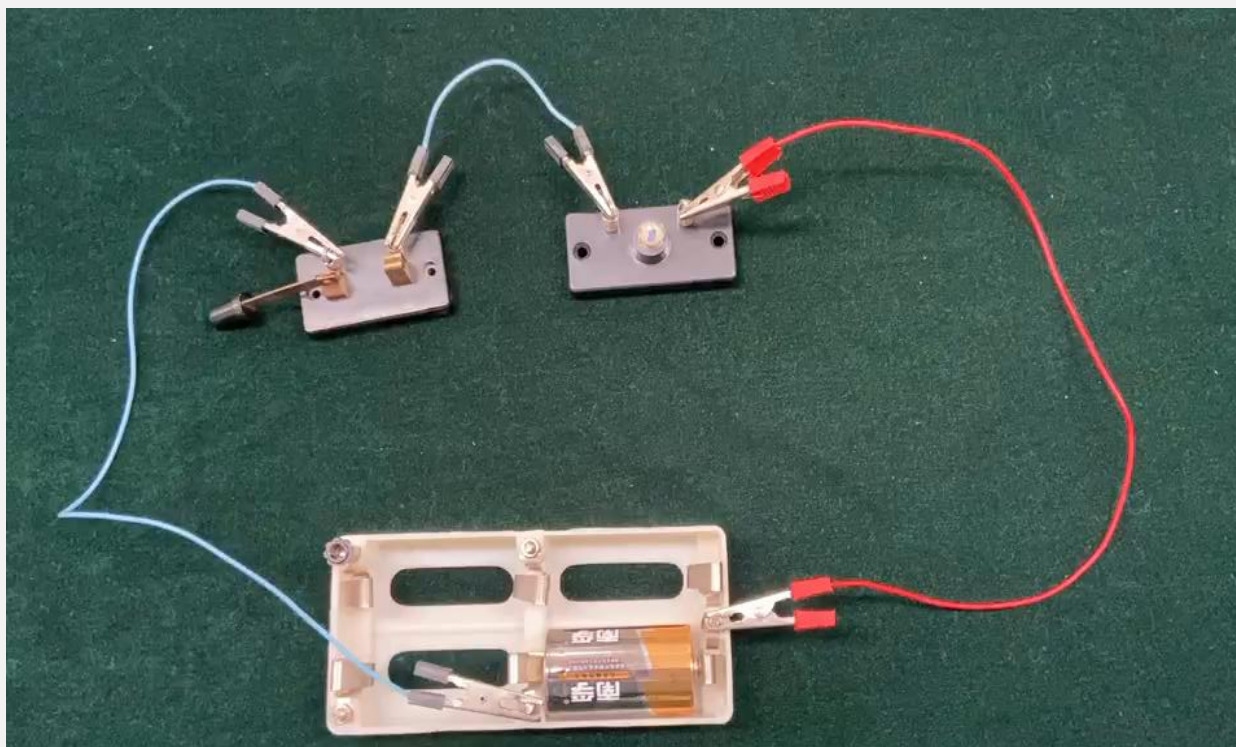
$$E_{\text{测}} = 1.5\text{V}$$

(量程为10V直流电压档)

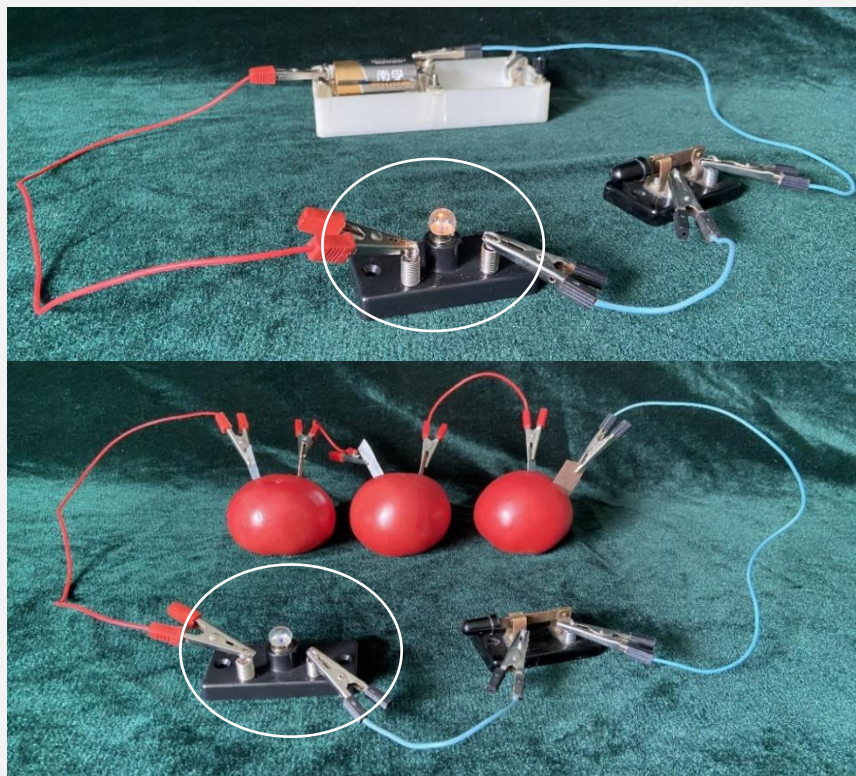


$$E_{\text{测}} = 2.5\text{V}$$

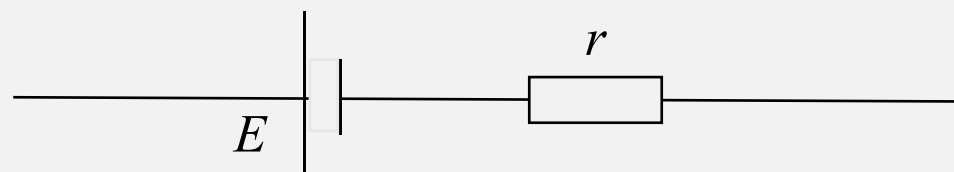
看一看，发生了什么？

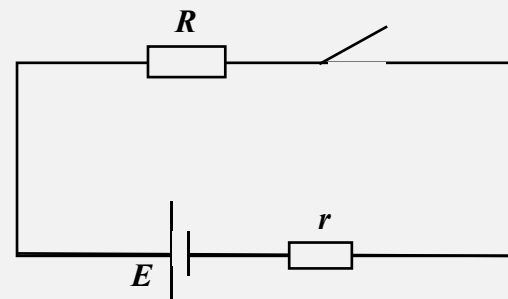
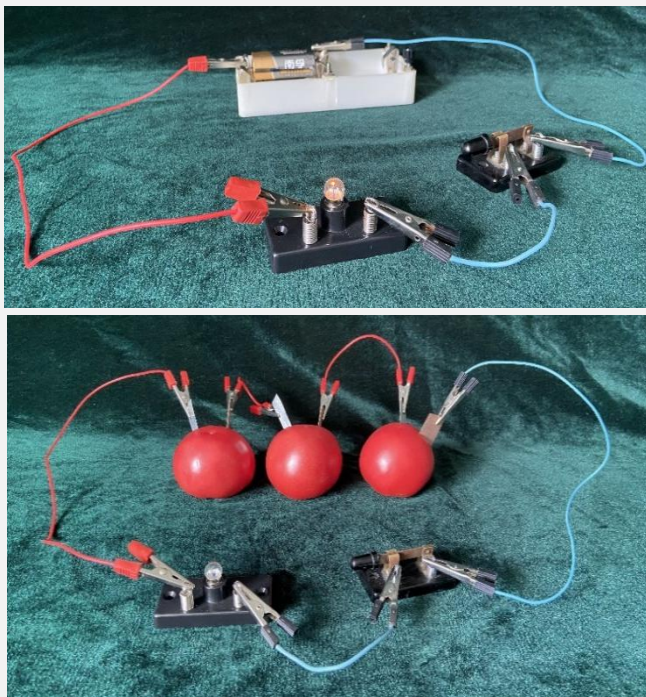


想一想



电池是如何
影响小灯泡
的亮暗的呢？

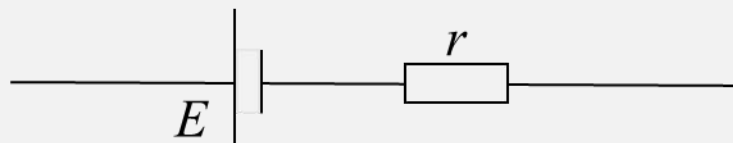




$$I = \frac{E}{R + r}$$



想一想

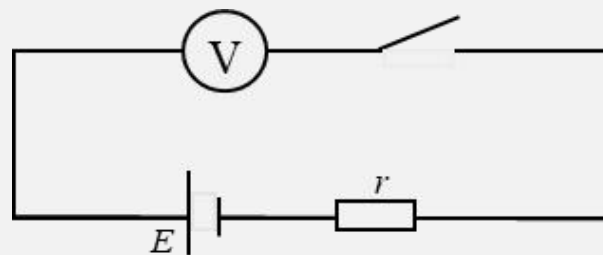


如何通过实验测量电池电动势和内阻呢？

$$U=E-Ir$$

$$I=0$$

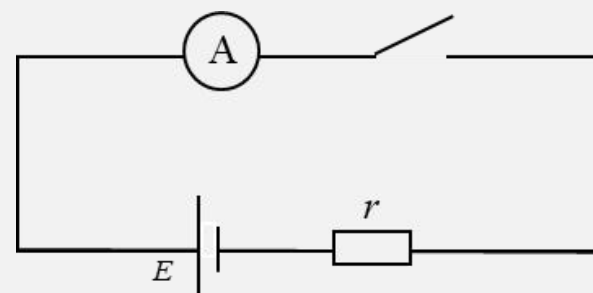
$$U_{\text{断}}=E$$



$$I = \frac{E}{R + r}$$

$$I_{\text{短}} = \frac{E}{r}$$

$$r = \frac{E}{I_{\text{短}}}$$





想一想

根据闭合电路的欧姆定律:

$$E=U_{\text{外}}+U_{\text{内}}$$

$$I=\frac{E}{R+r}$$

再配合着老师提供的器材,
你还能提出什么实验设计方案吗?

电压表、电流表、滑动变阻器、电池、开关



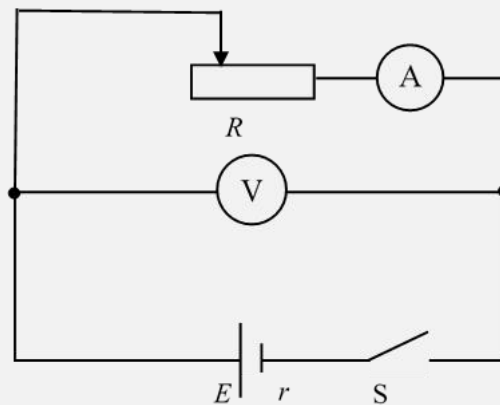
设计方案

$$E = U_{\text{外}} + U_{\text{内}}$$

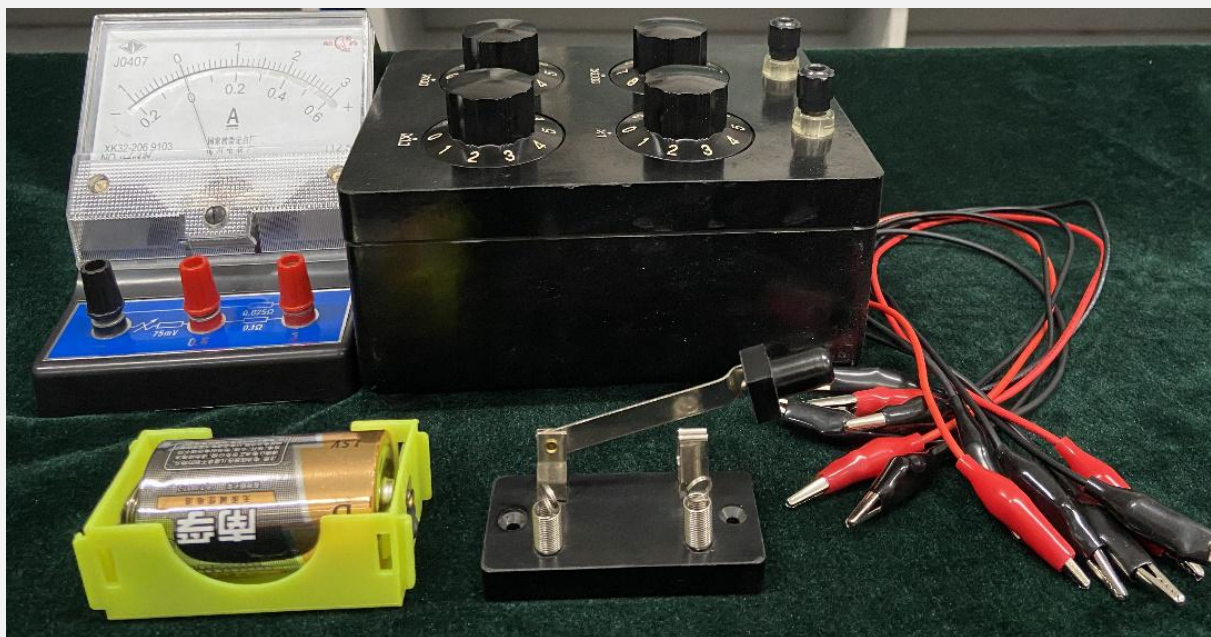
$$U_{\text{内}} = Ir$$

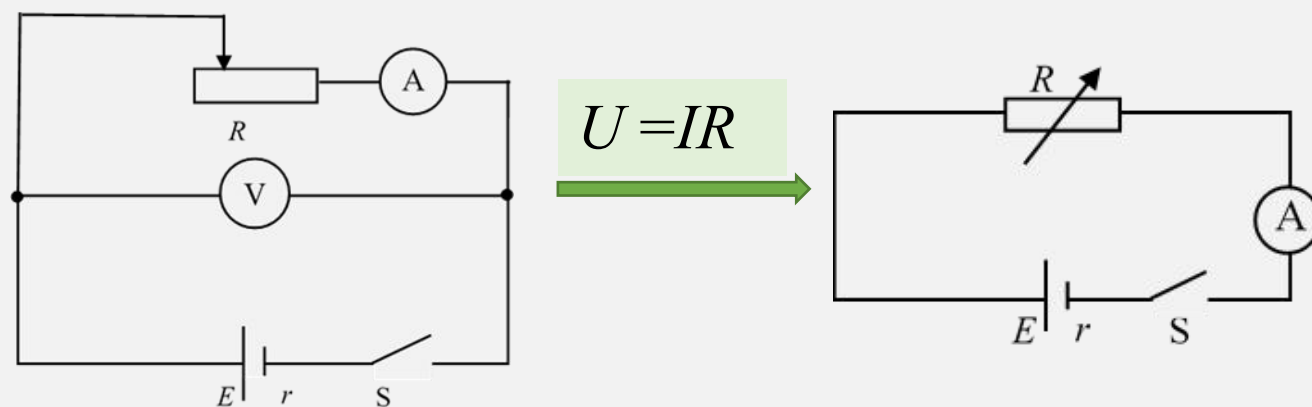
$$E = U + Ir$$

$$\begin{pmatrix} U_1 & I_1 \\ U_2 & I_2 \end{pmatrix}$$



电流表、电阻箱、电池、开关



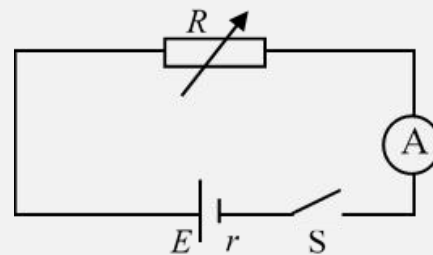


$$E=U+Ir$$

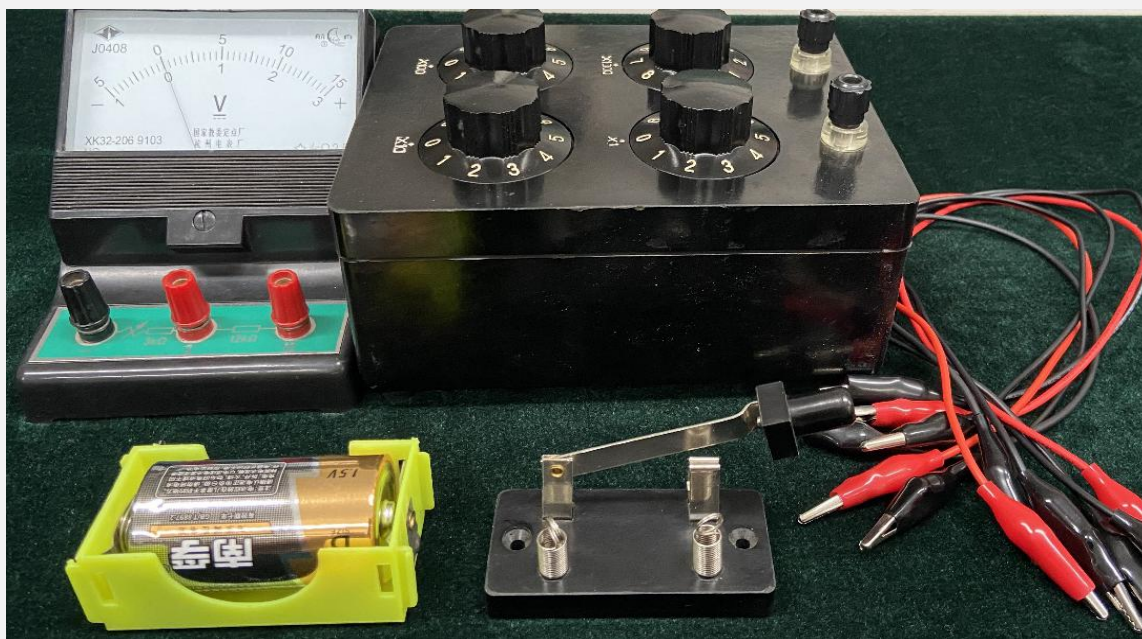
$$U=IR$$

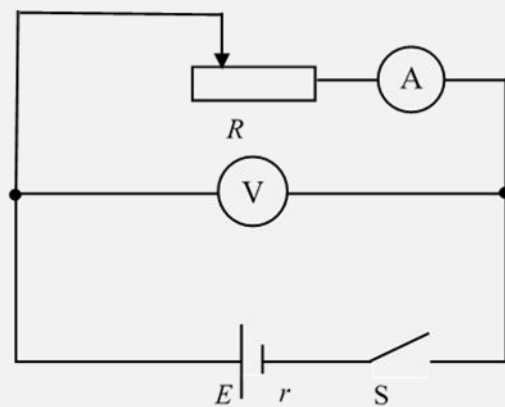
$$E=IR+Ir$$

$$\begin{pmatrix} I_1 & R_1 \\ I_2 & R_2 \end{pmatrix}$$

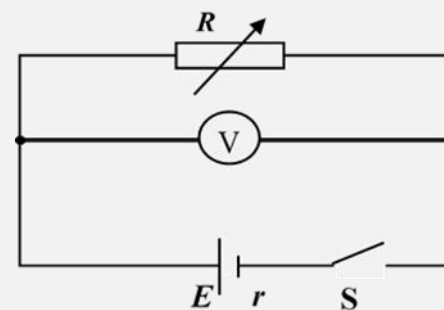


电压表、电阻箱、电池、开关





$$I = \frac{U}{R}$$



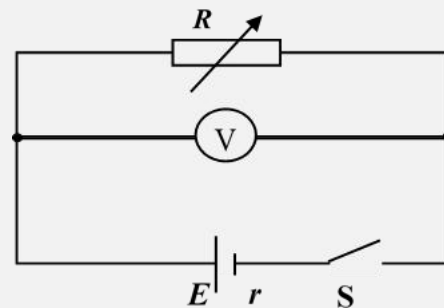
$$E=U+Ir$$

$$I = \frac{U}{R}$$

$$E = U + \frac{U}{R} r$$

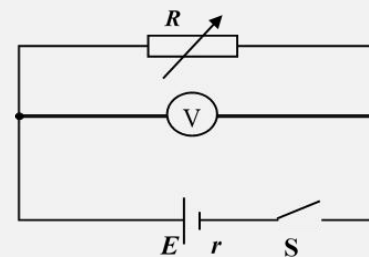
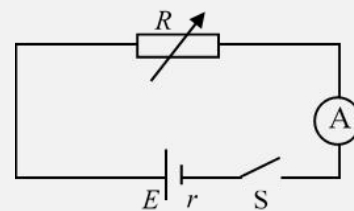
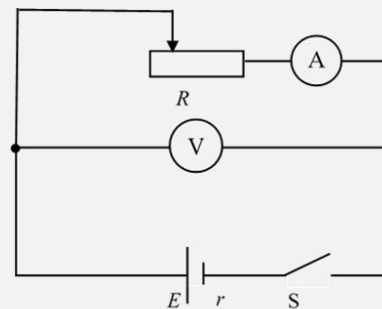
$$(U_1 \quad R_1)$$

$$(U_2 \quad R_2)$$



方案评估

$$E = U_{\text{外}} + U_{\text{内}} \left\{ \begin{array}{l} E = U + Ir \\ E = IR + Ir \quad (U = IR) \\ E = U + \frac{U}{R} r \quad (I = \frac{U}{R}) \end{array} \right.$$





想一想

在上述方法的基础上,如何减小因测量所带来的偶然误差呢?

多组数据取平均

$$E=U+Ir$$

$$\begin{pmatrix} U_1 & I_1 \\ U_2 & I_2 \end{pmatrix}$$

$$\begin{pmatrix} U_3 & I_3 \\ U_4 & I_4 \end{pmatrix}$$

$$\begin{pmatrix} U_5 & I_5 \\ U_6 & I_6 \end{pmatrix}$$



方案评估

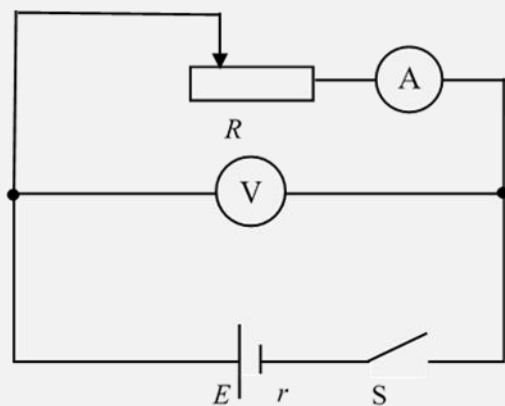
多组数据取平均的方法，误差减小了，
但处理起来比较复杂。



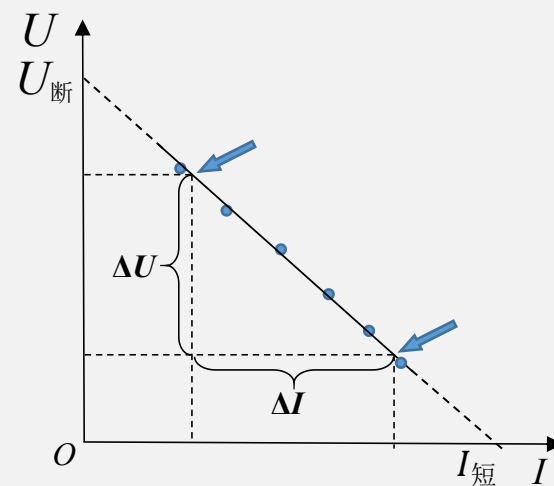
想一想

在上述方法的基础上，有没有更简便、直观的方法呢？

图像法



$$U = E - Ir$$



$$U_{\text{断}} = E$$

$$r = \frac{E}{I_{\text{短}}}$$

$$r = \left| \frac{\Delta U}{\Delta I} \right|$$



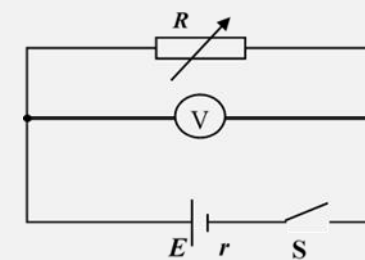
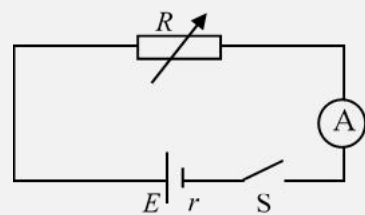
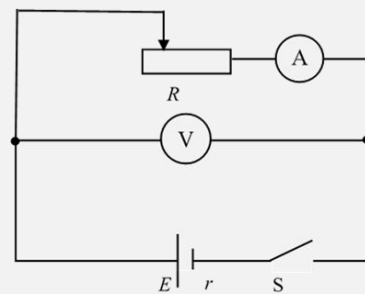
方案评估

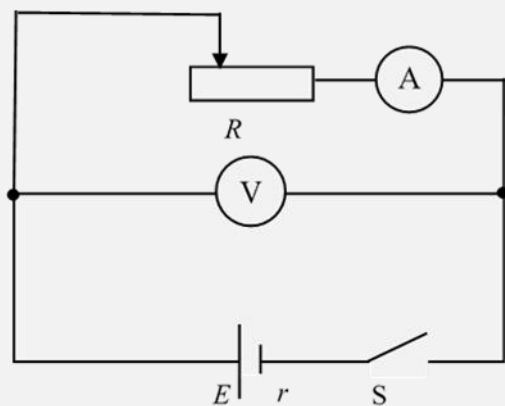
误差减小了，处理起来比较简便、直观。

小结

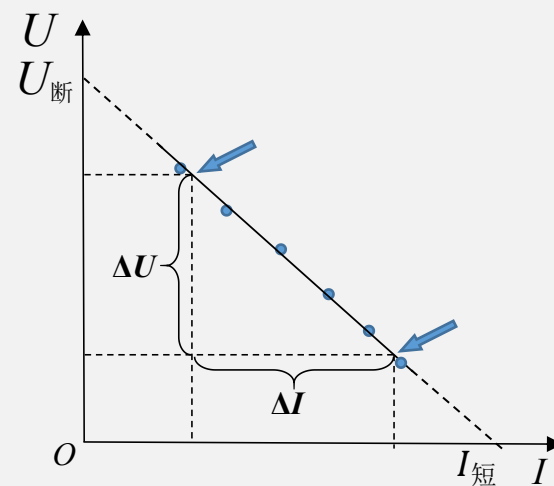


$$E = U_{\text{外}} + U_{\text{内}} \left\{ \begin{array}{l} E = U + Ir \\ E = IR + Ir \quad (U = IR) \\ E = U + \frac{U}{R}r \quad (I = \frac{U}{R}) \end{array} \right.$$





$$U = E - Ir$$

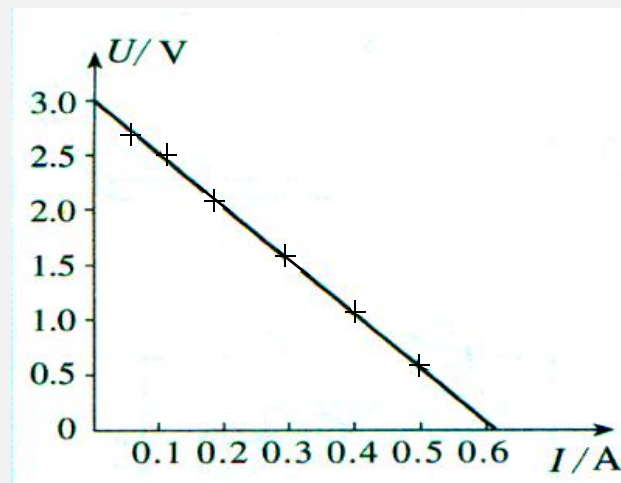
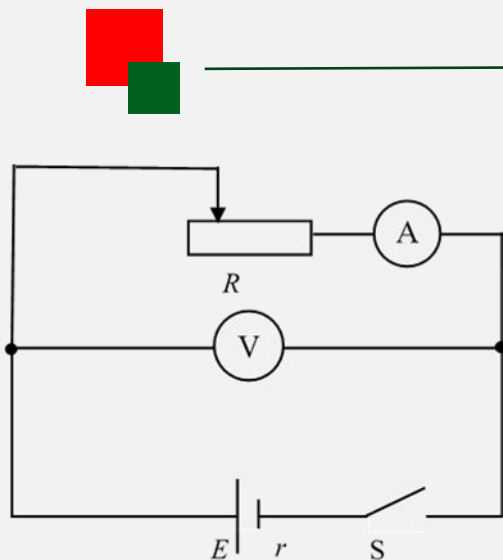


$$U_{\text{断}} = E$$

$$r = \frac{E}{I_{\text{短}}}$$

$$r = \left| \frac{\Delta U}{\Delta I} \right|$$

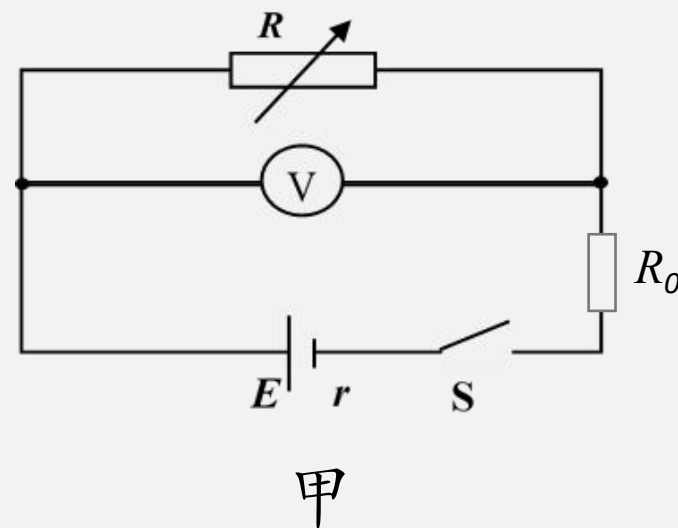
例题



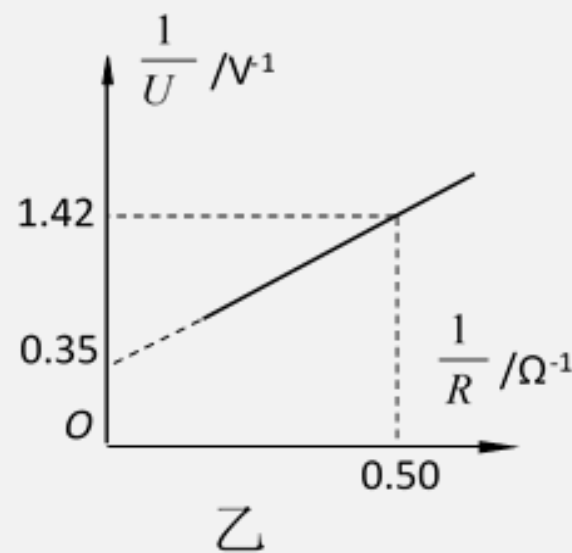
在如图所示的电路中，改变 R 的阻值，从电压表和电流表中读出几组 I 、 U 值，在坐标纸上以 I 为横坐标， U 为纵坐标，用测出的几组 I 、 U 值画出 $U-I$ 图像（如图所示）。可得电源电动势为3.0 V，电池内阻为5.0 Ω 。（保留两位有效数字）

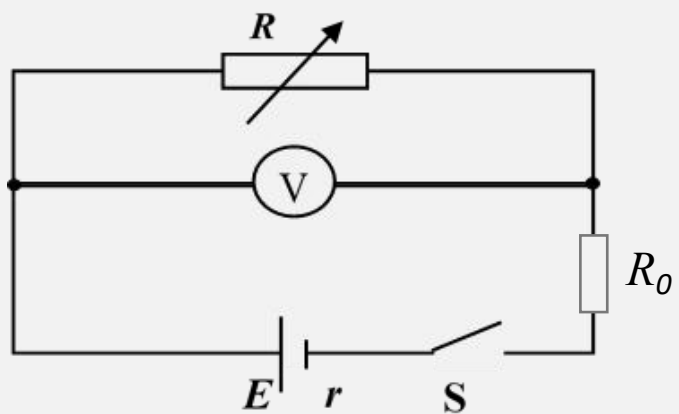
拓展训练

为了测量由两节干电池组成的电池组的电动势和内电阻，某同学设计了如图甲所示的实验电路，其中 R 为电阻箱， $R_0=5.0\Omega$ 为保护电阻。断开开关 S ，调整电阻箱的阻值，再闭合开关 S ，读取并记录电压表的示数及电阻箱接入电路中的阻值。多次重复上述操作，可得到多组电压值 U 及电阻值 R 。



并以 $\frac{1}{U}$ 为纵坐标，以 $\frac{1}{R}$ 为横坐标，画出 $\frac{1}{U} - \frac{1}{R}$ 的关系图线（该图线为一直线）如图乙所示。由图线可求得电池组的电动势 $E = \underline{\hspace{1cm}} \text{V}$ ，内阻 $r = \underline{\hspace{1cm}} \Omega$ 。（保留两位有效数字）



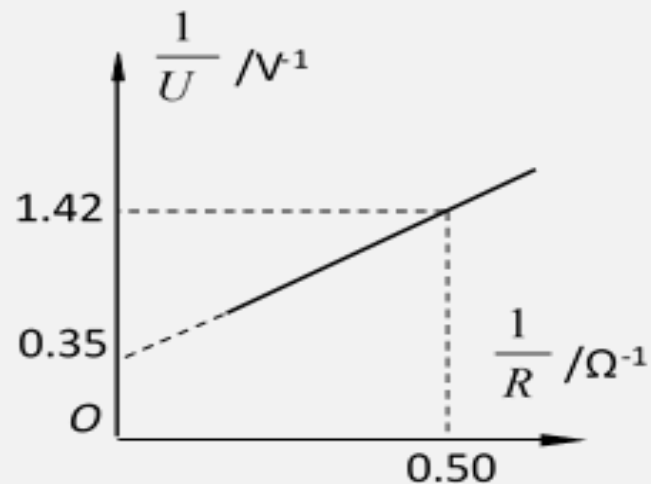


甲

$$U = IR$$

$$I = \frac{E}{R + r + R_0}$$

$$U = \frac{R}{R + R_0 + r} E$$



乙

由 $U = \frac{R}{R + R_0 + r} E \Rightarrow \frac{1}{U} = \frac{R_0 + r}{E} \cdot \frac{1}{R} + \frac{1}{E}$

可知图线的斜率 $k = \frac{R_0 + r}{E}$

纵截距为 $\frac{1}{E}$

由图线数据可得 $\frac{1}{E} = 0.35 \Rightarrow E = 2.9(\text{V})$

$$k = \frac{R_0 + r}{E} = \frac{1.42 - 0.35}{0.50} = 2.14 \Rightarrow r = 2.14E - R_0 = 2.14 \times \frac{1}{0.35} - 5.0 = 1.1(\Omega)$$



思考

在使用外电压和总电流的测量方案时，电压表和电流表的内阻对实验测量结果有影响吗？如果有，测得的电动势和内阻值与真实值之间的大小关系又如何呢？